ATAS
ADVANCED TRADING ANALYTICAL SOFTWARE

# COMPLETE GUIDE
# Reading Smart Tape

# Introduction

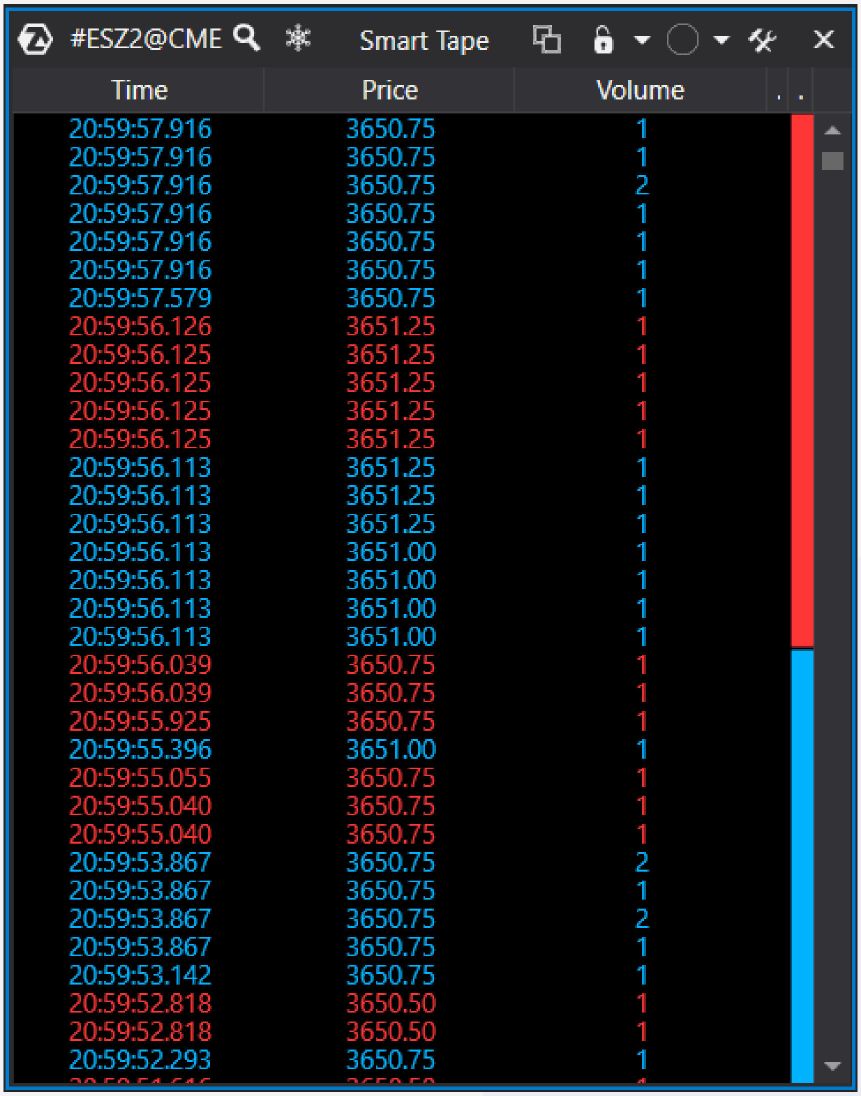
There are many methods and instruments that help traders analyze and predict price movements in the world of financial markets. Tape analysis is one of these methods. The tape is a stream of information about completed trades, displayed in real-time.

This article will explore various aspects of tape reading and its main characteristics. We will also study indicators that are based on the tape: Order Flow Indicator, Big Trades, and others. We will delve into the main settings and ways to apply these indicators for market analysis and trading decisions.

Let's start our fascinating journey into the world of Order Flow analysis and reveal all its secrets.

# What is a tape?

A tape is the primary volume analysis instrument. It displays the sequence of trades taking place on the exchange.

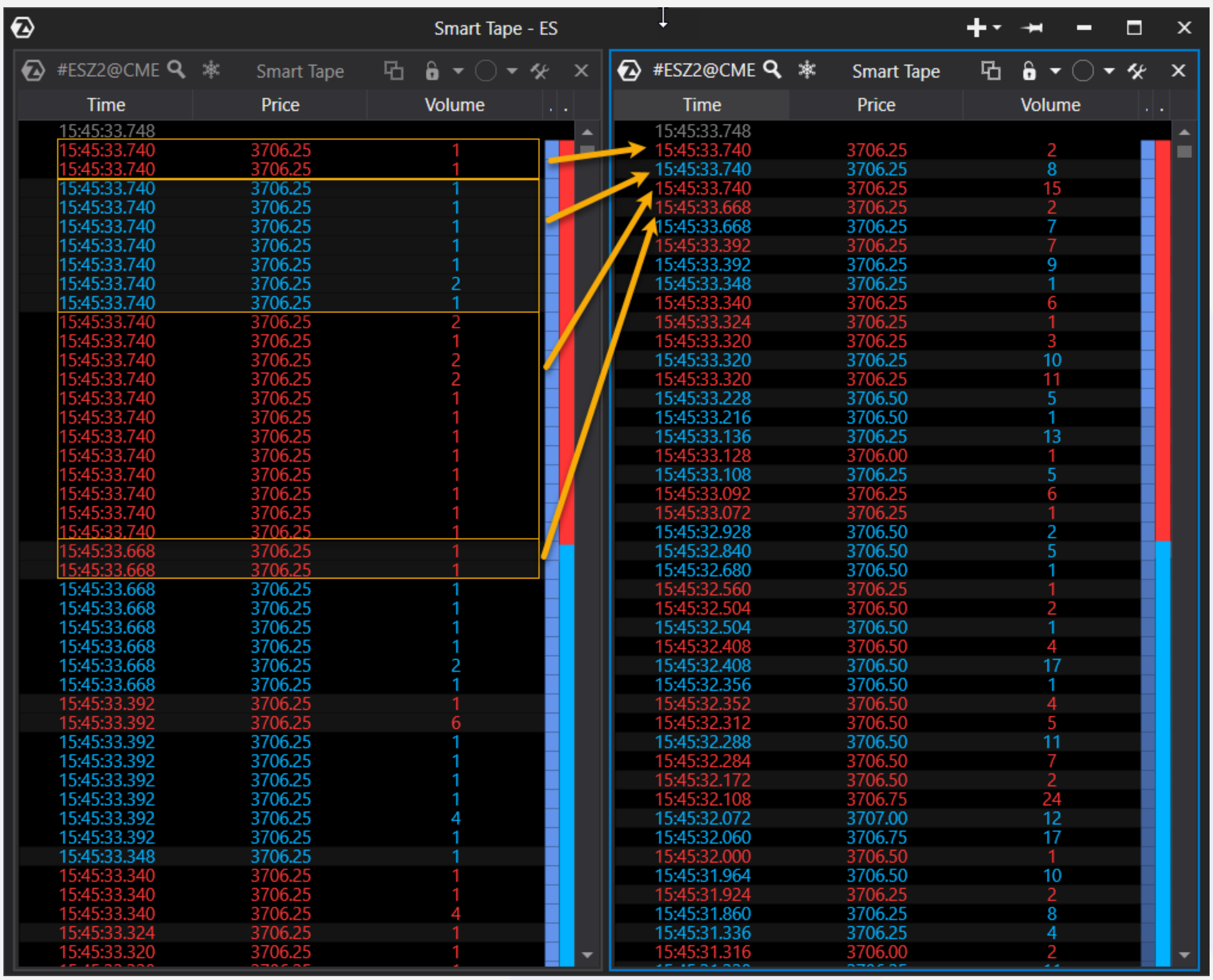



However, the tape in its classic form may be a bit useless and difficult to read since the data comes in fragmented.

Suppose there are 100 Buy Limit orders at the price of 1001 and another 100 Buy Limit orders at the cost of 1000 on the market, while the volume of each placed order is one lot. A big seller comes and sends a Sell Market order with a volume of 150 lots. As a result, this Sell Market order will be executed using a hundred orders for 1001 and fifty orders for 1000. Therefore, the large order from one player was completed using 150 orders from other market participants. As a result, you will see 100 lines of sales at the price of 1001 and 50 lines of sales at 1000 in the tape, all with a quantity of 1 lot each. You must admit that it can be pretty challenging to visually assess 150 lines of executed trades at once and discern the presence of a big seller behind them.

To solve this problem, a unique “Smart Tape” has been developed in the ATAS platform. It shows the real volumes of buys and sells of big players. The principle is simple: the Smart Tape aggregates individual trades so that the user sees a real market order that was originally placed. In the example above, the Smart Tape would show only one line with a volume of 150.

The screenshot below shows two tapes: on the left - a classic one, and on the right - an aggregated one.



The process of order flow analysis is called “tape reading”, “Order Flow analysis”, or simply “order flow reading”. The flow of trades, particularly aggregated ones, served as the basis for a range of Order Flow indicators. This method allows you to focus on finding large aggregated trades, analyzing flow speed, and identifying Order Flow patterns to determine potential traces of trading robots.

What is essential to analyze in the context of Order Flow analysis?

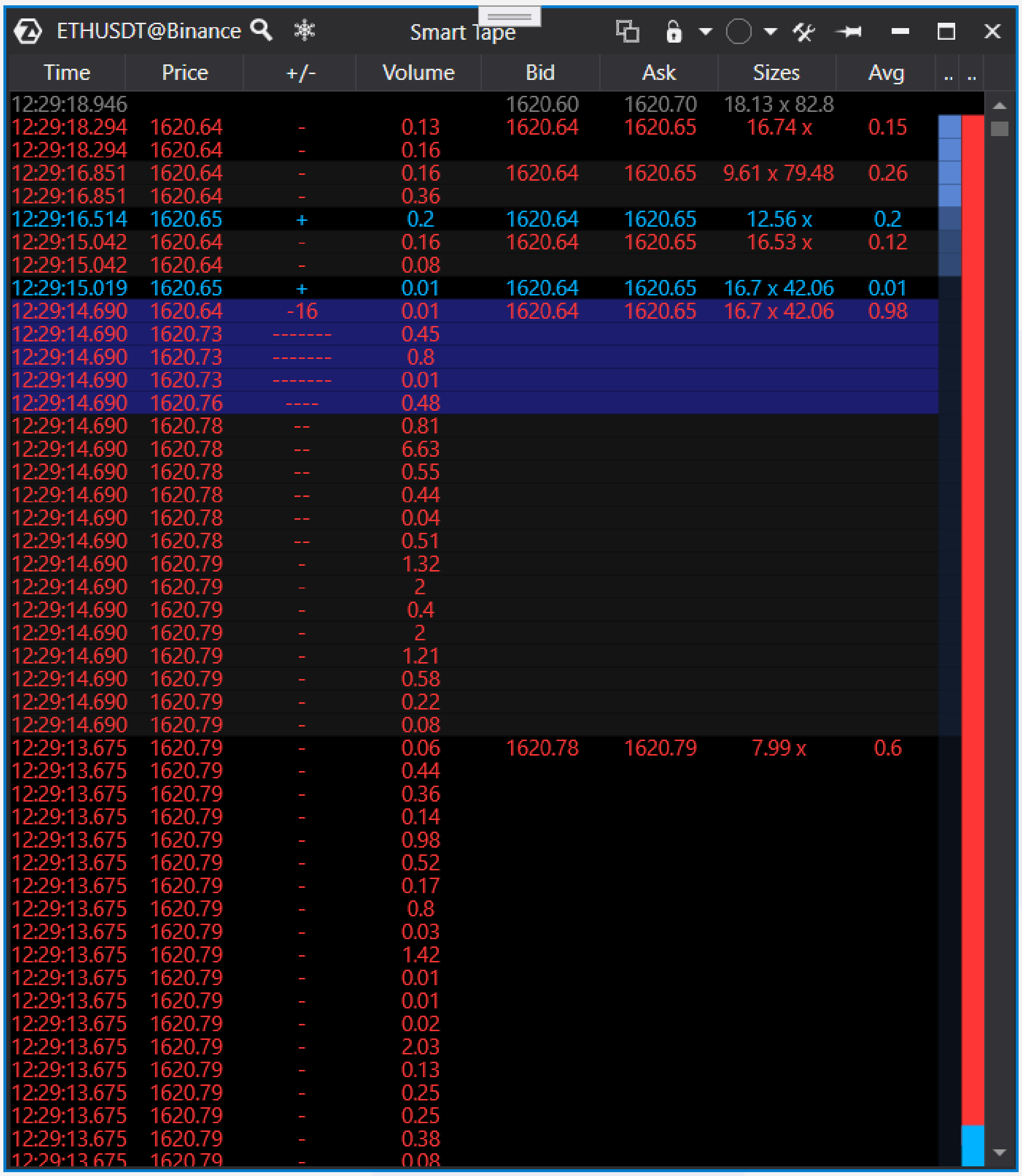
- Aggregated print volumes — to identify large market orders.
- Large volumes of non-aggregated trades indicate the presence of significant funds on both the aggressive and passive sides.
- The speed of trades allows you to understand what phase the market is in.
- The volumes of the Best Bid and Best Ask provide insight into the strength of resistance from passive market participants.
- Iceberg orders.
- Places where stop orders are activated.

# Smart Tape

## Basic settings and application

The "Smart" prefix in the name of the ATAS tape originated from implementing our proprietary algorithm that aggregates individual trades into original orders.

Let's look at the list of available data in the tape, starting with the classical, unaggregated (fragmented) mode.

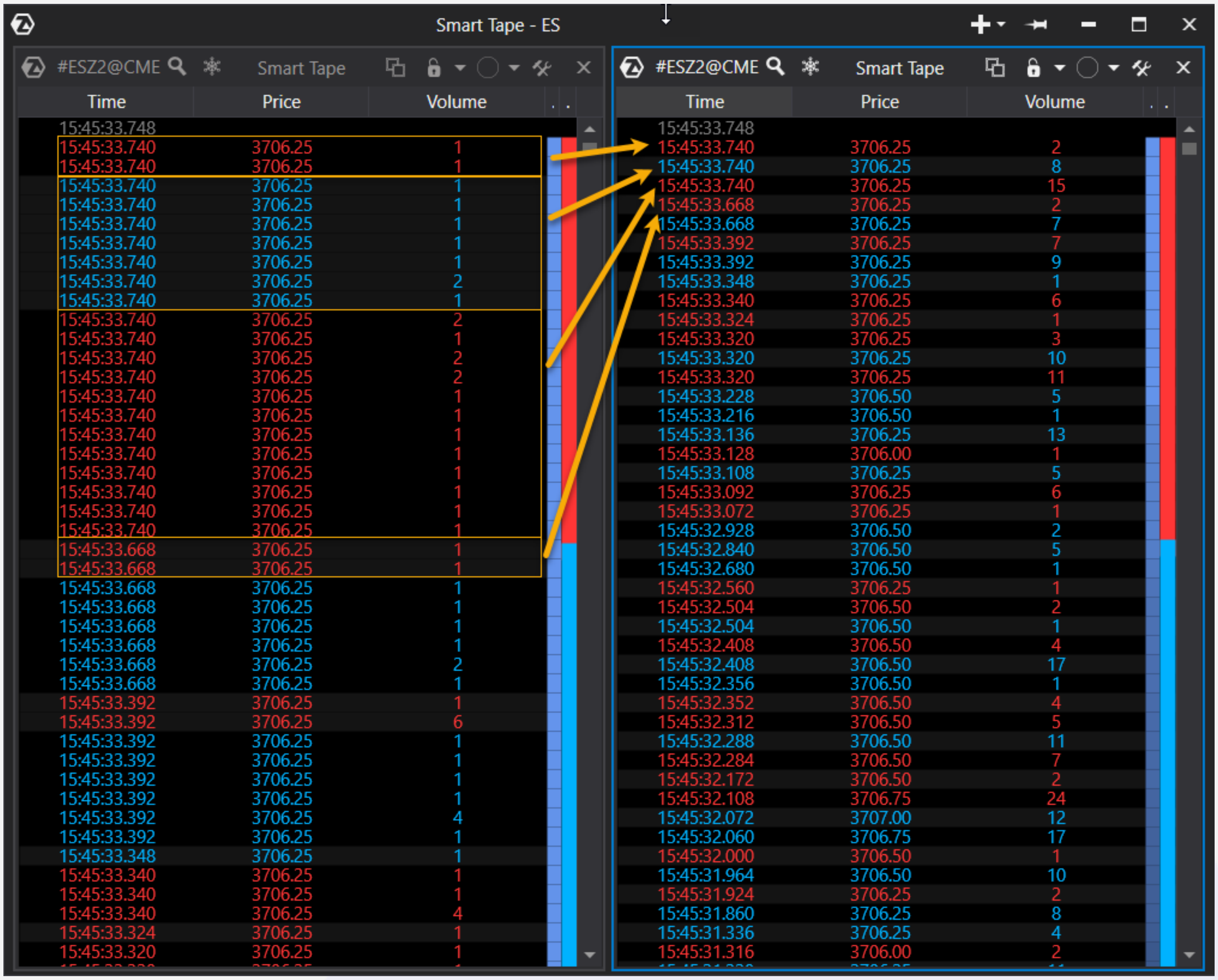



| Time | Price | +/- | Volume | Bid | Ask | Sizes | Avg |
|---|---|---|---|---|---|---|---|
| 12:29:18.946 | | | | 1620.60 | 1620.70 | 18.13 x 82.8 | |
| 12:29:18.294 | 1620.64 | - | 0.13 | 1620.64 | 1620.65 | 16.74 x | 0.15 |
| 12:29:18.294 | 1620.64 | - | 0.16 | | | | |
| 12:29:16.851 | 1620.64 | - | 0.16 | 1620.64 | 1620.65 | 9.61 x 79.48 | 0.26 |
| 12:29:16.851 | 1620.64 | - | 0.36 | | | | |
| 12:29:16.514 | 1620.65 | + | 0.2 | 1620.64 | 1620.65 | 12.56 x | 0.2 |
| 12:29:15.042 | 1620.64 | - | 0.16 | 1620.64 | 1620.65 | 16.53 x | 0.12 |
| 12:29:15.042 | 1620.64 | - | 0.08 | | | | |
| 12:29:15.019 | 1620.65 | + | 0.01 | 1620.64 | 1620.65 | 16.7 x 42.06 | 0.01 |
| 12:29:14.690 | 1620.64 | -16 | 0.01 | 1620.64 | 1620.65 | 16.7 x 42.06 | 0.98 |
| 12:29:14.690 | 1620.73 | ------- | 0.45 | | | | |
| 12:29:14.690 | 1620.73 | ------- | 0.8 | | | | |
| 12:29:14.690 | 1620.73 | ------- | 0.01 | | | | |
| 12:29:14.690 | 1620.76 | ---- | 0.48 | | | | |
| 12:29:14.690 | 1620.78 | -- | 0.81 | | | | |
| 12:29:14.690 | 1620.78 | -- | 6.63 | | | | |
| 12:29:14.690 | 1620.78 | -- | 0.55 | | | | |
| 12:29:14.690 | 1620.78 | -- | 0.44 | | | | |
| 12:29:14.690 | 1620.78 | -- | 0.04 | | | | |
| 12:29:14.690 | 1620.78 | -- | 0.51 | | | | |
| 12:29:14.690 | 1620.79 | - | 1.32 | | | | |
| 12:29:14.690 | 1620.79 | - | 2 | | | | |
| 12:29:14.690 | 1620.79 | - | 0.4 | | | | |
| 12:29:14.690 | 1620.79 | - | 2 | | | | |
| 12:29:14.690 | 1620.79 | - | 1.21 | | | | |
| 12:29:14.690 | 1620.79 | - | 0.58 | | | | |
| 12:29:14.690 | 1620.79 | - | 0.22 | | | | |
| 12:29:14.690 | 1620.79 | - | 0.08 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.06 | 1620.78 | 1620.79 | 7.99 x | 0.6 |
| 12:29:13.675 | 1620.79 | - | 0.44 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.36 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.14 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.98 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.52 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.17 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.8 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.03 | | | | |
| 12:29:13.675 | 1620.79 | - | 1.42 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.01 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.01 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.02 | | | | |
| 12:29:13.675 | 1620.79 | - | 2.03 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.13 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.25 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.25 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.38 | | | | |
| 12:29:13.675 | 1620.79 | - | 0.08 | | | | |

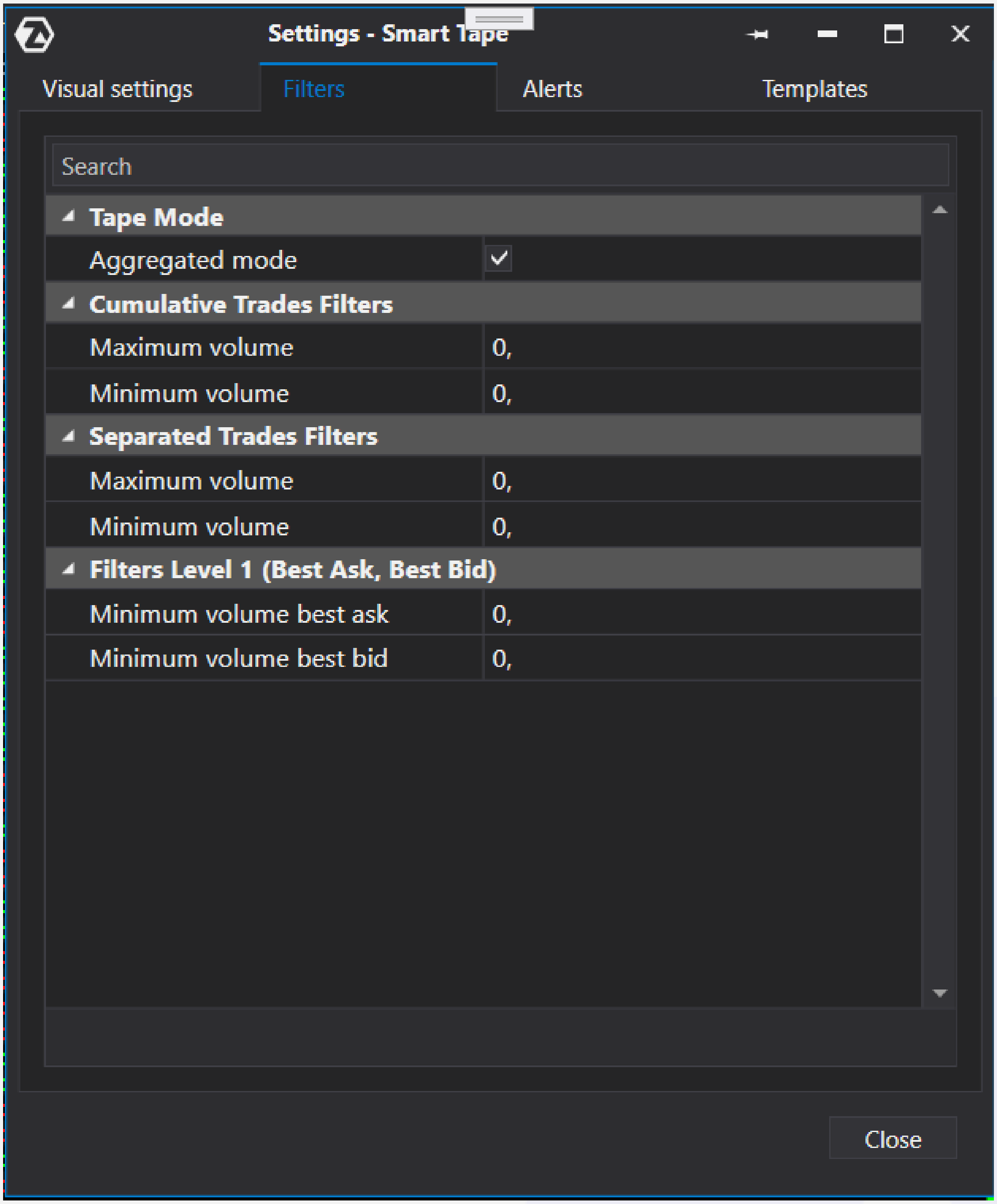
- Time — the exact moment when trade took place;
- Price — the trade's price;
- +/- — a trade's direction indicator shows the number of consecutive Up/Down ticks. Let's take the price of $100 as a starting point. A trade occurred at $101 - this is an UpTick, and we will see a "+". Then another trade occurred at $101 — “+” remains. If the next print occurs at $102, we will see ++. If a trade with a price of $101 comes after $102, we will see “-”;
- Volume — the trade volume;
- Bid/Ask and Sizes — best bid/ask prices and their volumes after completing the aggregated trade. By enabling the “Show previous level 1 data” mode in the settings, you can additionally track the Best Bid/Ask before the start of an aggregated trade. This information can help identify Iceberg orders. For example, the volume of the Best Bid at a price of $100 was 900 lots, and there were 10 trades executed at this price, each with a volume of 50. In total, a volume of 500 lots was traded. It is logical to expect that the volume of 400 Lot should remain at the Best Bid. If the volume of the Best Bid after the completion of the aggregated trade becomes, let's say, 600, it means that 200 lots were executed using a buy Iceberg order.
- Avg - the average size of the fragmented trade volume within the aggregated one. Example: 10 trades of 100 lots each are aggregated into one aggregated trade. Avg of such a trade will be 100;
- The tape speed indicator (blue vertical area) shows the speed of the tape at the moment;
- The buyers/sellers strength indicator shows the dominance of buyers or sellers within the last twenty seconds.

To switch Smart Tape to aggregation mode, go to settings ➔ Filters ➔ Aggregated mode.

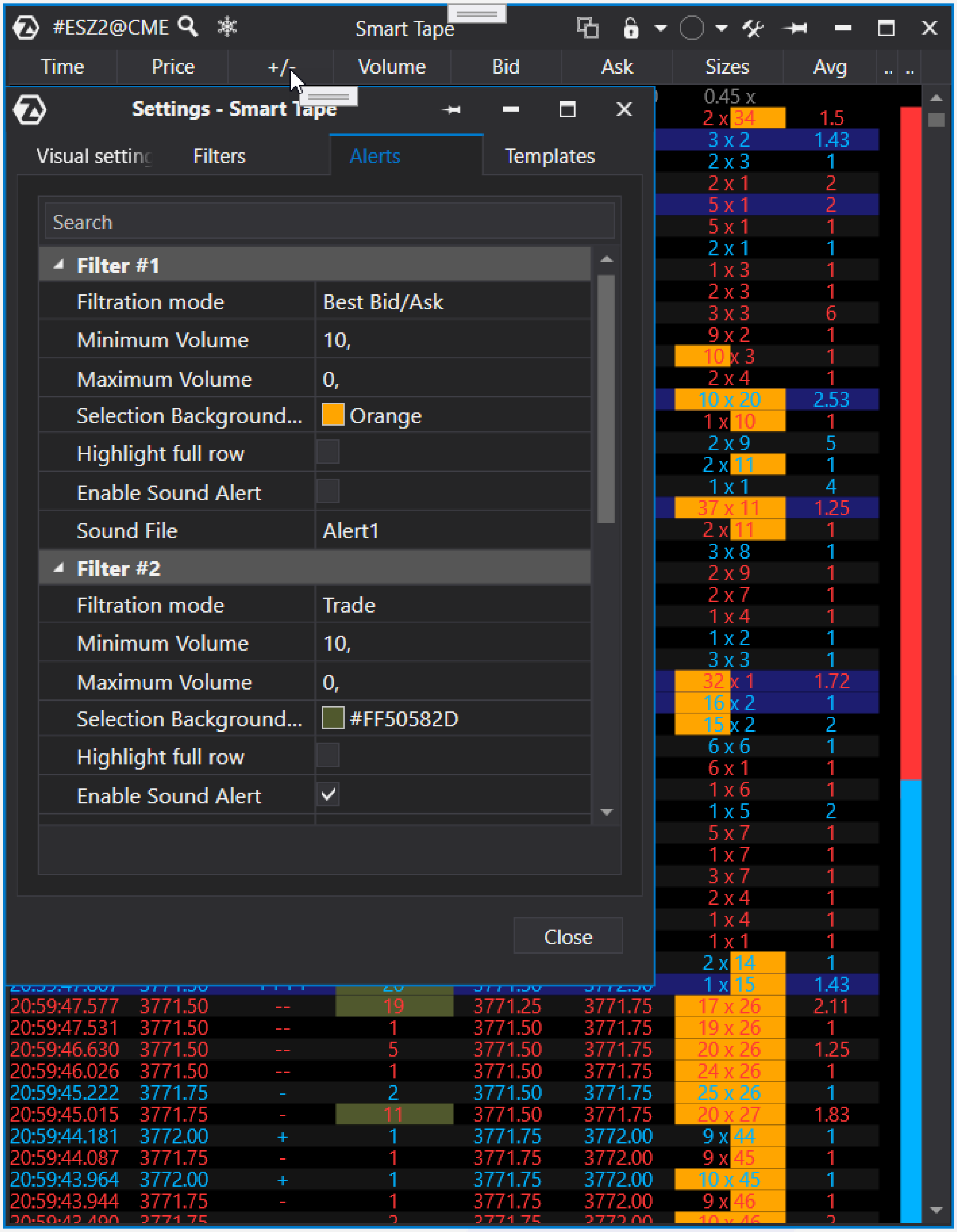
Let's use a screenshot already familiar to you to visualize the same data: on the left, we see fragmented trades, and on the right - aggregated ones:




The filtering system allows you to obtain trade selections based on aggregated or fragmented volumes, as well as based on the volume of the Best Bid/Ask.




If you need to highlight (without hiding) only the necessary trades, you can use the "Alerts" section. It enables you to customize the highlighting conditions flexibly and, if necessary, notifications about new trades that meet the filters.


#ESZ2@CME
Smart Tape
Time
Price
+/-
Volume
Bid
Ask
Sizes
Avg
Settings - Smart Tape
Visual settings
Filters
Alerts
Templates
Search
Filter #1
Filtration mode
Best Bid/Ask
Minimum Volume
10,
Maximum Volume
0,
Selection Background...
Orange
Highlight full row
Enable Sound Alert
Sound File
Alert1
Filter #2
Filtration mode
Trade
Minimum Volume
10,
Maximum Volume
0,
Selection Background...
#FF50582D
Highlight full row
Enable Sound Alert
Close

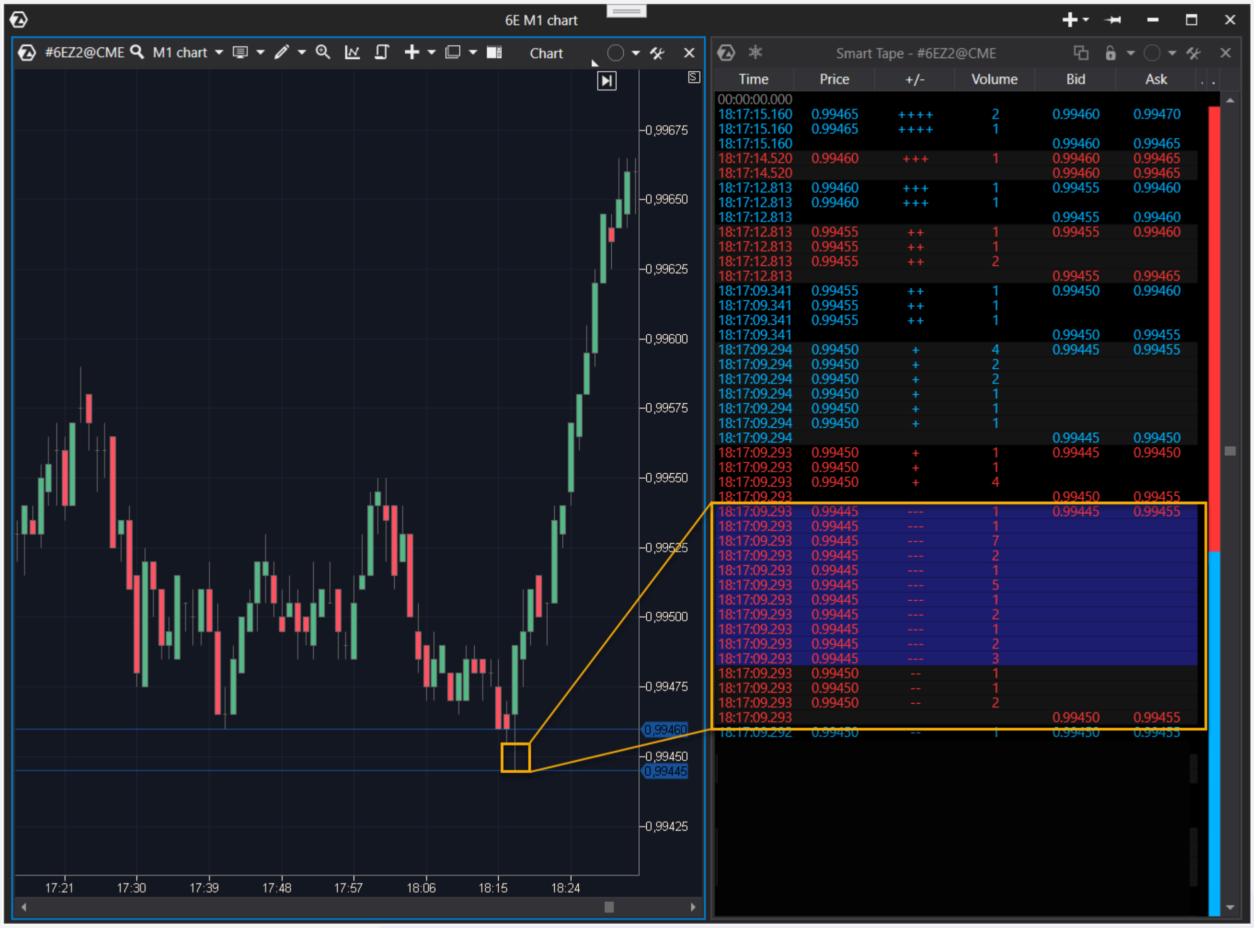
## Slippage and determination of stop order activation levels

**Slippage** is the execution of a market order at a price worse than the Best Bid (for selling) or Best Ask (for buying).

Let's look at an example of how it works. Best Bid at $100 has a volume of 10 lots. The next sell limit level at $99 has a volume of 5 lots. If in this situation a market seller comes into the market and submits a Sell Market order with a volume of 12 lots, then 10 of them will be executed at a price of $100, while the remaining 2 lots will be executed for $99. This situation is called slippage. Slippage usually occurs when large market orders are submitted or when stop orders are triggered.

Smart Tape allows you to highlight such moments with the “Above Ask and Below Bid” option.

This option particularly highlights places where stop orders are likely to be triggered. Many traders use the same trading strategies and technical analysis instruments. And place their stop orders at approximately the same price levels. When the price reaches stop order levels, brokers aggregate the stops into larger market orders and send them to the exchange. As a result, we see large aggregated market buy/sell orders with significant slippage in the tape.




In the picture, you can see a highlighted aggregated trade and its constituent fragmented trades. The candlestick chart shows that the previous Low was at the 0.99460 level. As we can see from the tape, as soon as a trade was made at 0.99450 (slightly below the previous Low), it must have triggered a stop order with slippage from 0.99450 to 0.99445 (5 ticks). Unfortunately, it is impossible to determine whether this stop order belonged to an individual trader or a group.

It is believed that significant funds manipulating the market know where the stop orders of the crowd are concentrated. The manipulation strategy in this case involves pushing the price to the levels of stop orders, placing limit orders beneath these stops (in the case of buying), getting the stops triggered, and getting your orders executed at the most favorable prices.

## Smart Tape. Conclusions

Aggregated tape is a handy feature. It enables you to see real market orders. However, it is not enough to analyze only the size and direction of orders. A big market buy can be executed using an even larger limit sell, an Iceberg order, or even a stop order of a crowd.

When a significant aggregated transaction appears, it usually makes sense to analyze it:

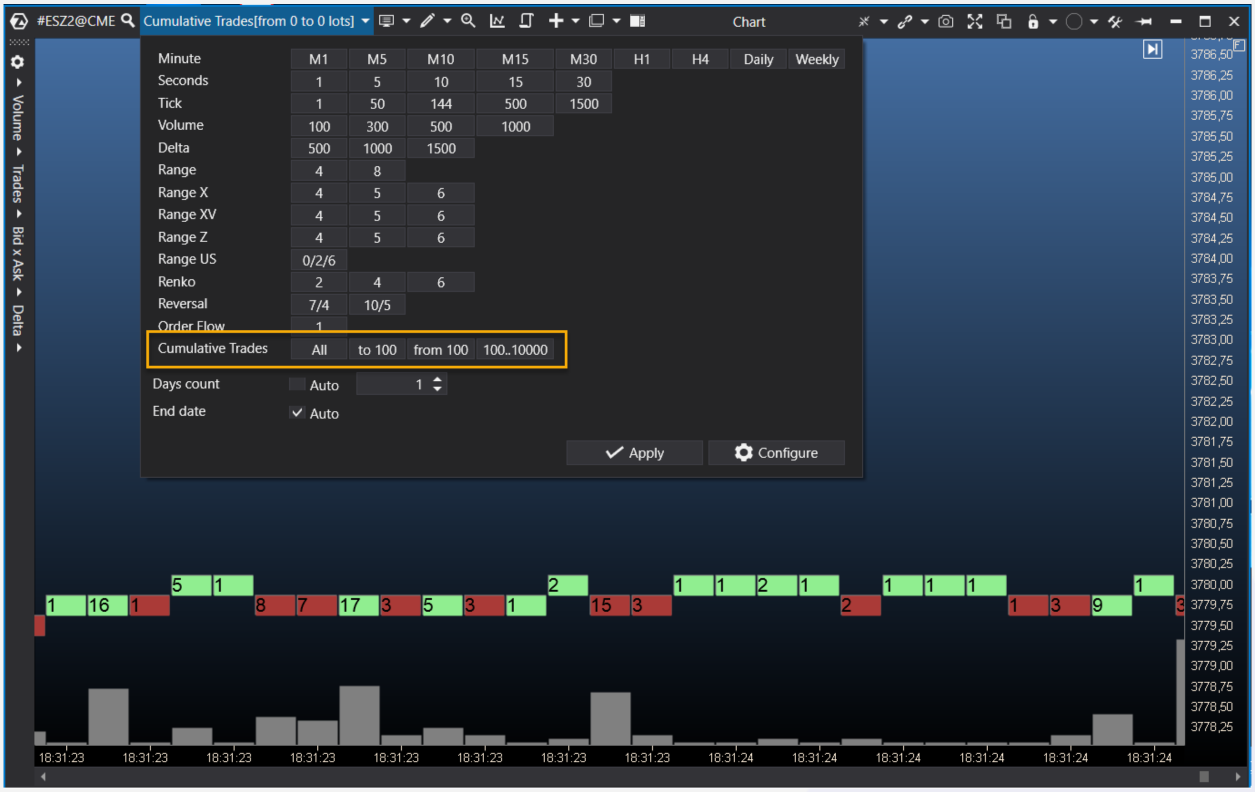
- if it consists of multiple small fragmented trades, we understand that on one side of this trade, there is a big participant, and on the other side, there is a crowd;
- if the volume of the aggregated trade is large and it consists of just a few comparable fragmented trades in terms of volume, it indicates that there is also a big player on the passive (limit) side;
- if a part of such trade was consumed by an Iceberg order, it is worth considering the intentions of the party who placed that Iceberg order;
- if an aggregated trade occurs with significant slippage, while updating the local High or Low, it is highly likely that a crowd's stop order triggered it.

Timeframe: Cumulative Trades

# Timeframe: Cumulative Trades

Smart Tape and aggregation provide a lot of helpful information. However, only some find reading data flow in a tabular format convenient.

To solve this problem, a specific timeframe was created - Cumulative Trades.




The volume range of aggregated trades can be set in the timeframe settings. For example, you can display trades of 100 lots and above. Each aggregated trade is visualized as a rectangle with a given color.

By compressing the chart horizontally, you can see a broader picture with greater height:


In addition to a separate timeframe, a number of specialized indicators for the chart have been developed. They will be discussed below.

# Order Flow Indicator

The indicator displays the flow of trades on top of a regular bar chart. It enables you to see the current flow of transactions simultaneously with the bar chart of any timeframe.


The indicator displays both regular fragmented trades and aggregated ones.

Filters allow you to highlight only large volumes and hide trades with volumes less than the specified ones, eliminating unnecessary noise.

## Big Trades

The indicator displays large aggregate buys and sells directly on the chart.


By default, the indicator automatically selects filters so only the maximum volumes are displayed on the current chart. If desired, you can configure arbitrary filters by volume.

Time filters allow you to customize the display only during certain trading hours. It is no secret that the working hours of different exchanges and trading activities can vary greatly. Time filters enable you to adjust volumes for each trading session flexibly.

One of the advantages of this indicator is the ability to display the slippage of an aggregated trade.


In this picture, the aggregated trade is highlighted with a circle, and the vertical rectangle with prices starting from 3789.00 to 3792.50 displays the price range where the individual trades within this aggregated trade took place.

Remember that slippage often shows places where stop orders are triggered.

# Adaptive Big Trades

An indicator without any filters. The idea is the same as Big Trades, but only 10 maximum trades are displayed in the currently visible range. If you zoom in or narrow the chart, the scale of the indicator changes dynamically.

This indicator is handy when you need to look in depth in detail.

# Tape patterns

A very flexible indicator that allows you to find chains of trades by a variety of criteria:

- session time;
- the volume of fragmented trades within the chain;
- the volume of aggregated trades within the chain;
- the number of trades within the chain;
- the range of ticks where all trades of the chain must be located;
- type of trades (buys, sells, intra-spread trades);
- time filter in milliseconds, which works in two modes:
    - looking for chains of trades within a given time interval. For example, if the filter value is 1000ms (equivalent to 1 second), the indicator will search for chains of trades within one second, meaning the time of the last trade cannot exceed the time of the first trade plus one second;
    - looking for trade chains with a time interval between them that does not exceed the specified parameter. For example, you can find a number of trades so that the time between each pair ranges from 5 to 10 ms.

In general, the indicator is versatile and multifunctional. It was created primarily to search for various patterns and traces of algorithms, but it can also be used in other situations.


For instance, in the section dedicated to Smart Tape, we mentioned that significant fragmented trades within an aggregated trade indicate a contradictory situation. On one side, there is a big aggressive buyer or seller whose order was executed by matching it with an equally substantial passive order.

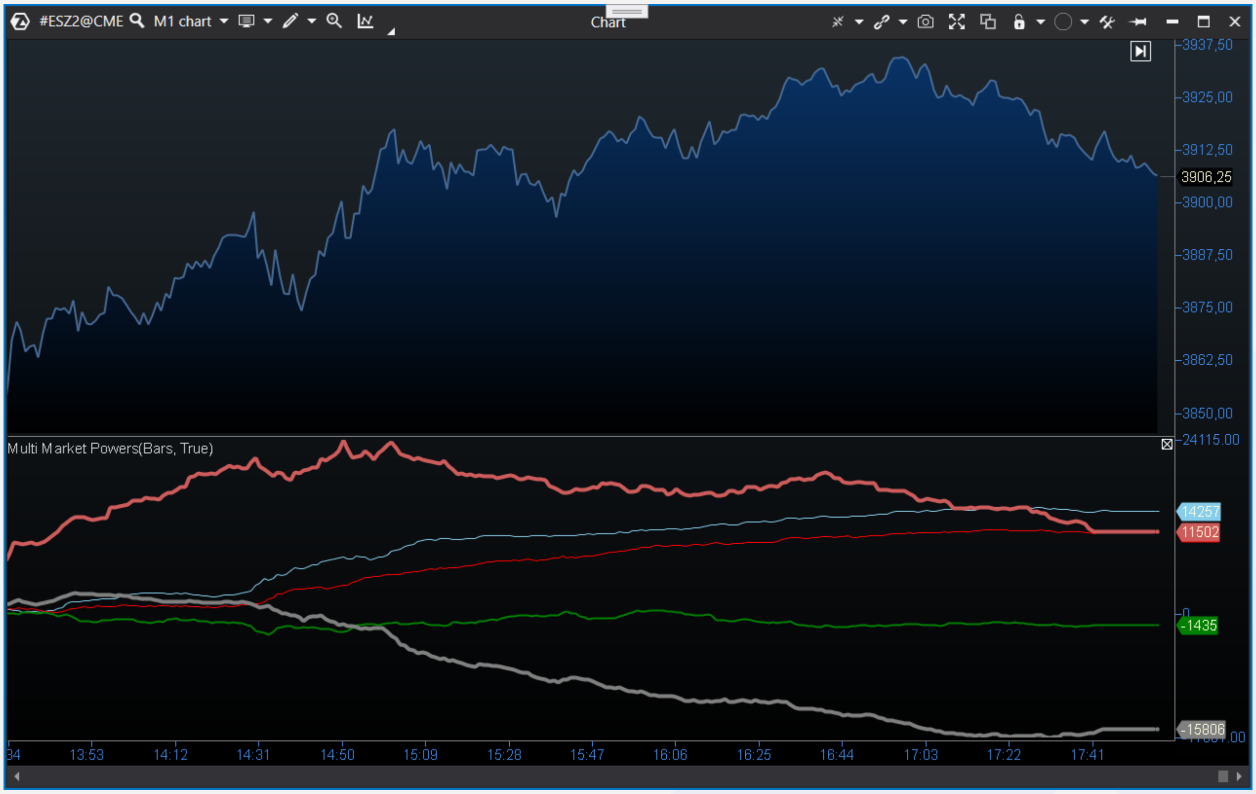
With the help of the Tape Patterns indicator, you can easily filter such situations by adjusting the range of volumes for fragmented trades within aggregated ones.

# CVD pro, CVD pro (multi)

The purpose of these indicators is to show the correlation (or lack of it) between the direction of trades of a certain volume and the price.

In CVD Pro, you can specify a range of trade volumes. The indicator will only consider trades within that specified range and generate a curve representing the difference between market buys and sells. For example, you can see the dynamics of the difference between buyers and sellers with volumes of 50 lots and more.

The CVD pro (multi) indicator was developed to track several volume ranges at the same time. It shows the change in the delta of different volume groups at once.




In this example, the red line represents the cumulative delta (difference between market buys and sells) of trades above 40 lots. The price correlation is immediately noticeable. At the same time, the gray line represents the delta of small volumes. Here the correlation is the opposite. It is an excellent illustration of how those trading with large market orders move in sync with the market while those trading with small orders move in the opposite direction (losing out).

It is essential to understand that instruments can have different characteristics. It is important not to rely on guaranteed profits in the blind pursuit of big trades. Each tool requires researching, identifying volume drivers, and paying attention to trading times (different market sessions may have different players, algorithms, etc.).

## Speed of Tape

As the name implies, the indicator was created to analyze the data speed in the tape. The main parameter of the indicator is the period, which is set to a default value of 15 seconds. The indicator measures the average tape speed within this period and visualizes it on the chart by highlighting bars with above-average speed.

# Conclusion

Tape analysis is an essential instrument for traders. It helps to make informed decisions in financial markets. However, like any other analysis method, it requires practice and experience to achieve success.

We hope this article has helped you understand the basics of tape analysis and inspired you to explore this fascinating trading area further.

Wishing you success on your journey as a trader!